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*2044/21970*

Printed Pages : 7 TEE-303

(Following Paper ID and Roll No. to be filled in your Answer Book)

PAPER ID : 2044 Roll No.

# B. Tech.

## (SEM. III) EXAMINATION, 2007-08

## NETWORK ANALYSIS & SYNTHESIS

*Time : 3 Hours]* *[Total Marks : 100*

*Note :* *(1) Attempt **all** questions.*

*(2) All questions carry **equal** marks.*

*(3) In case of numerical problems assume data wherever not provided.*

1 Attempt any **four** parts of the following : **5×4=20**

(a) Draw the oriented graph of a network with fundamental outset matrix as given below :

| *Twigs* | | | | *Links* | | |
|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| 1 | 0 | 0 | 0 | -1 | 0 | 0 |
| 0 | 1 | 0 | 0 | 1 | 0 | 1 |
| 0 | 0 | 1 | 0 | 0 | 1 | 1 |
| 0 | 0 | 0 | 1 | 0 | 1 | 0 |

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(b) A resistive network is given in **Fig. 1(b)**. Obtain the loop incidence matrix.

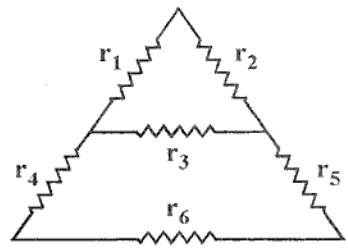



**Fig. 1(b)**

(c) Derive the generalised KCL equation in terms of fundamental cut set matrix.

(d) Differentiate between Twig matrix and Link matrix with suitable example.

(e) A network is given in **Fig. 1(e)**. Obtain the loop currents using branch impedance matrix and Tie-set matrix.

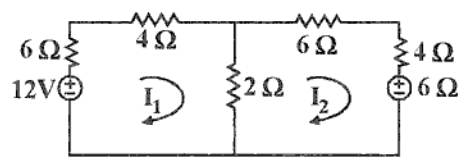



**Fig. 1(e)**

(f) Draw the dual networks of the circuits shown in **Fig. 1f(1)** and **(2)**

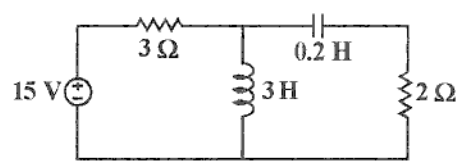



**Fig. 1f(1)**


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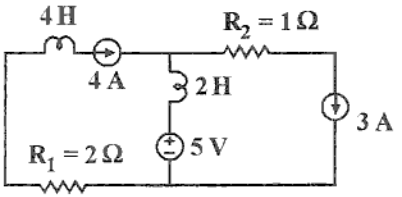



Fig. 1f (2)

2 Attempt any **three** parts of the following : $6\frac{2}{3} \times 3 = 20$

(a) Using superposition theorem, find the voltages across $(4 + j3)\Omega$ in the network shown in **Fig. 2(a)**.

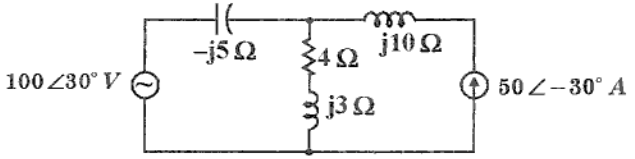



Fig. 2(a)

(b) Obtain Thevenin's equivalent of the network across terminals AB for the circuit shown in **Fig. 2(b)**.

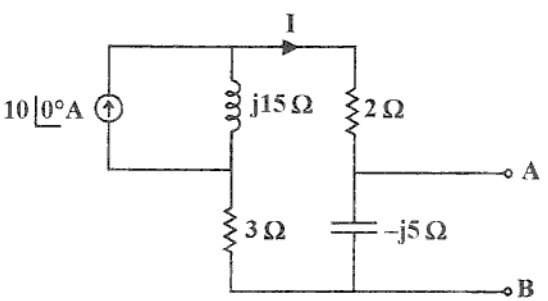



Fig. 2(b)


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(c) In the network shown in **Fig. 2(c)**, find the value of $R_L$ which results in maximum power transfer.

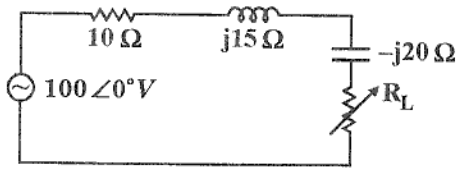



**Fig. 2(c)**

(d) Verify the Tellegen's theorem for the network shown in **Fig. 2(d)**.

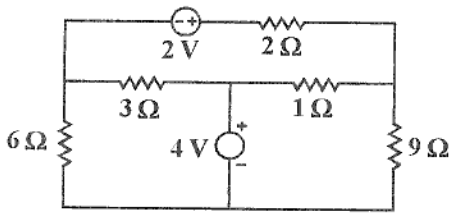



**Fig. 2(d)**

(e) Verify reciprocity theorem for $V$ and $i$ for the network shown in **Fig. 2(e).**

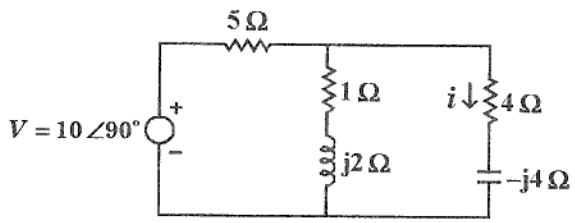



**Fig. 2(e)**



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3 Attempt any **two** parts of the following : **10×2=20**

(a) Obtain $V_2 / V_1$ of the network shown in **Fig. 3(a)**. Also find pole-zero configuration.

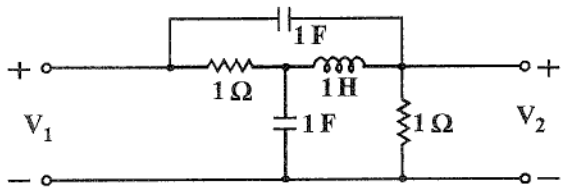



**Fig. 3(a)**

(b) Draw the Bode plot of the following transfer function

$$G(s) = \frac{1}{s(1+0.5\,s)\,(1+0.05\,s)}$$

(c) Enlist the properties of driving point function. Also check the stability criteria of the polynomial by applying Routh-Hurwitz criterion in

$$P(s) = s^6 + s^5 + 3s^4 + 3s^3 + 3s^2 + 2s + 1$$

Attempt any **two** parts of the following : **10×2=20**

(a) Obtain the Y-parameters of the network shown in **Fig. 4(a)**.

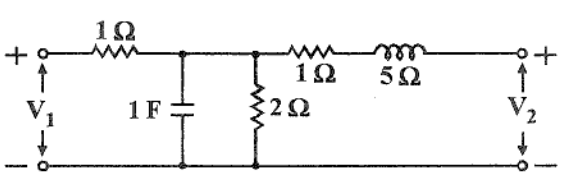



**Fig. 4(a)**

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(b) For a network the equations are

$$I_1 = 0.5\,V_1 - 0.2\,V_2$$

$$I_2 = -0.2\,V_1 + V_2$$

Find $Y$-parameters and ABCD parameters. Also find its equivalent $\pi$-network.

(c) Determine input and output impedances of a generalised network in terms of hybrid parameters.

5 Attempt any **two** parts of the following : **10×2=20**

(a) Design an m-derived high pass filter having a design impedance of 600 Ω, cut-off frequency of 5 kHz and m = 0.35. Also determine the frequency of infinite attenuation.

(b) Enlist the properties of RL driving point immittance function. Check whether

$$P(s) = 4s^6 + 2s^5 + 17s^4 + 8s^3 + 16s^2 + 6s + 3$$

is Hurwitz.

(c) (1) Check whether the function

$$Z(s) = \frac{2s^2 + 2s + 1}{s^3 + 2s^2 + s + 2}$$ is a positive real function.


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(2) The driving point impedance of an LC network is given by $Z(s) = \frac{10s^4 + 12s^2 + 1}{2s(s^2 + 1)}$.

Obtain the first form of Cauer LC network.


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